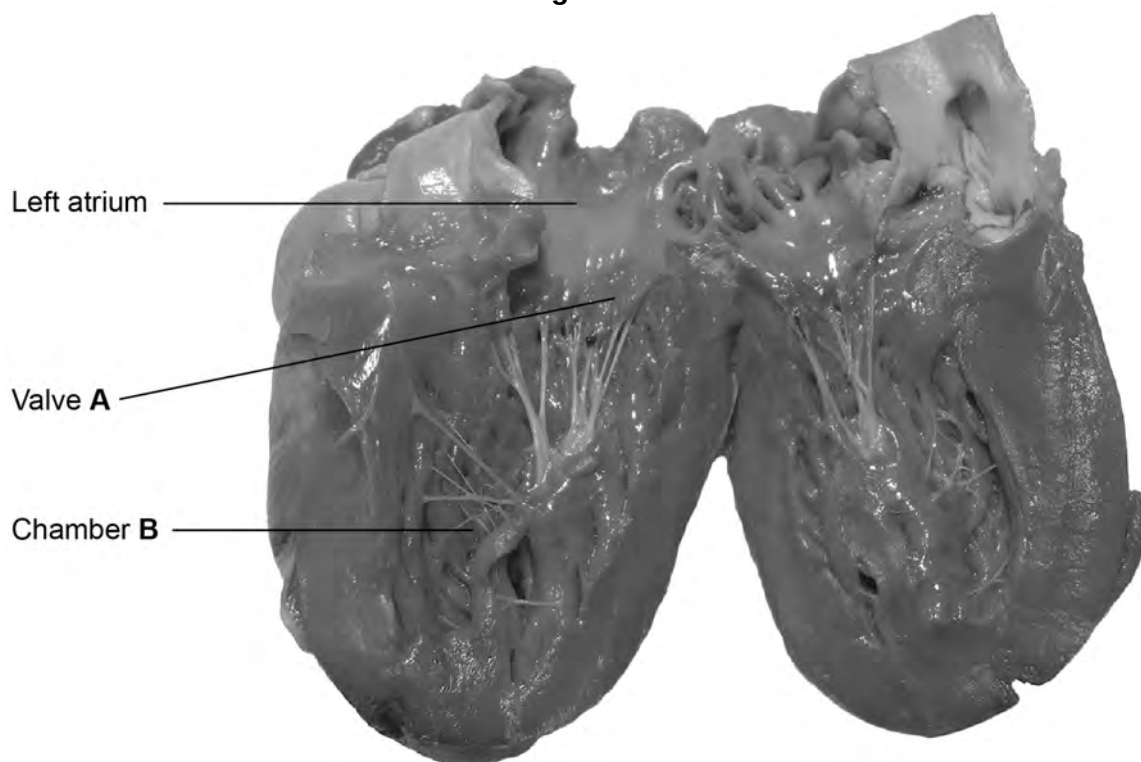


03

Figure 2 shows a photograph of a dissected heart.

Figure 2



03

1

Name valve **A** and chamber **B**.

[1 mark]

Valve **A** _____

Chamber **B** _____

03

2

Give **two** safety precautions that should be followed when dissecting a heart.

[1 mark]

1 _____

2 _____

0 3 . 3

Explain how valve **A** in **Figure 2** maintains a unidirectional flow of blood.

[2 marks]

Question 3 continues on the next page

A research scientist investigated the effect of caffeine on heart rate in human volunteers.

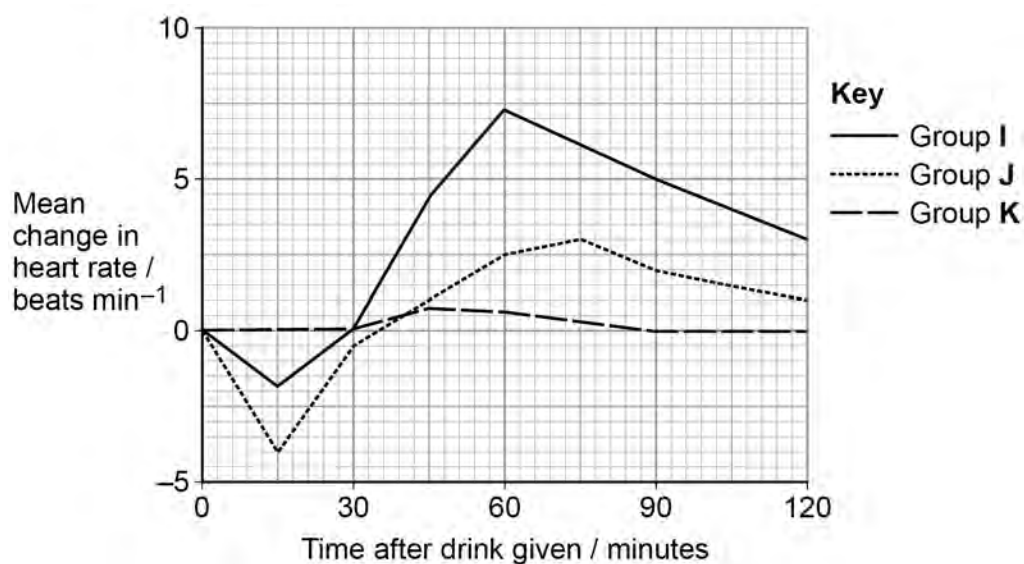
The scientist divided volunteers into three groups. Each group was given the same volume of fluid.

- Each member of Group **I** was given a sports drink containing caffeine and sugar.
- Each member of Group **J** was given a sports drink containing caffeine and no sugar.
- Each member of Group **K** was given water.

The scientist recorded the volunteers' heart rate before the drink was given and for 120 minutes after the drink was given.

Her results can be seen in **Figure 3**.

Figure 3



0 3 . 4 Caffeine affects the autonomic nervous system.

Suggest how caffeine could account for the results of Group **I** in **Figure 3** at 60 minutes.

[2 marks]

0 3 . 5 Before taking the drink, the mean heart rate of Group **J** was 68 beats per minute.

Fifteen minutes after taking the drink, the mean volume of blood leaving the hearts of Group **J** was 4700 cm^3 per minute.

Calculate the mean volume of blood leaving the heart at each beat fifteen minutes after taking the drink.

[1 mark]

Answer = _____ cm^3

Question 3 continues on the next page

0 3 . 6

The increase seen in Group I could be due to the combination of caffeine and sugar.

Suggest **one** drink to be given to an **additional** group that should be investigated to find out if this is true.

Give a reason for your answer.

[2 marks]

Group to be given _____

Reason _____

E

Turn over for the next question

*Do not
write
outside
the
border*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►
